Wednesday Warm Up: Units 3 and 4, Magnetic Forces and Fields, Induction

Prof. Jordan C. Hanson

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1 Memory Bank

- $V_{\rm H} = Blv$ ... The Hall voltage, given external B-field, height of conductor, l, and the drift velocity v.
- $\vec{F} = I\vec{L} \times \vec{B}$ ... The Lorentz force on a current.
- $\vec{F} = I\vec{L} \times \vec{B}$ ... Lorentz Force on a Current
- $\vec{F} = q\vec{v} \times \vec{B}$ ... Lorentz Force on a Charge
- $\vec{B} = (\mu_0 I)/(2\pi r)\hat{\phi}$ ... The magnetic field B *caused* by the current I a distance r away. The field *circles* the wire, so the direction of $\vec{B}$ is $\hat{\phi}$.
- $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.
- $\vec{B} = \mu_0 n I \hat{z}$ ... The B-field inside a solenoid. $n = N/L$, the turns per unit length.
- $\epsilon = -N\Delta\phi_m/\Delta t$ ... Faraday's Law
- $\phi_m = \vec{B} \cdot \vec{A}$ ... Definition of magnetic flux

2 The Hall Effect

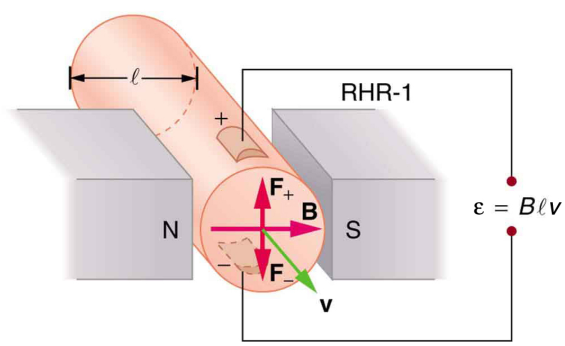

Figure 1: Measuring the Hall voltage.

1. A Hall probe is used to measure fluid flow (Fig. 1). Suppose a tube with diameter 1 cm carries a positively charged fluid moving at velocity v, in a 0.8 T B-field. The Hall voltage is measured to be 0.5×10^{-6} V. What is the fluid velocity?

2. What would happen to the Hall voltage if the charge carriers in the fluid were *negative* instead of *positive*?

3 Ampère's Law

1. Inside a solenoid, the B-field is constant, $\mu_0 n I$, where n is the number of wire coils per unit length. (a) If we measure B to be 1200 μT inside a solenoid running 0.5 A of current, what will the B-field be at 1 A? (b) If we cut the solenoid in half, what will the B-field be at 1 A?

4 Magnetic Induction: Faraday's Law

1. Suppose a uniform B-field of 0.01 T made an angle of 45 degrees with an wire loop with an area of 0.1 m^2. (a) Calculate ϕ_m, the magnetic flux. (b) Suppose the B-field drops to 0 T in 10 ms. What is $\Delta\phi_m/\Delta t$?

2. (a) An MRI technician moves her hand from a region of very low magnetic field strength into an MRI scanner's 2.00 T field with her fingers pointing in the direction of the field. Find the average emf induced in her wedding ring, given a diameter of 2.20 cm and assuming it takes 0.250 s to move it into the field. Would the current significantly raise the temperature of the ring?